

Dept. of Computer and Communication Engineering
Patuakhali Science and Technology University
6th Semester (Level-3, Semester-II), Midterm Examination of B.Sc. Engg. (CSE), July-December: 2021
Course Code: CCE-321 Course Title: Computer Peripheral and Interfacing
Credit Hour: 3.0 Full Marks: 15 Duration: 60 Minutes

1. ~~a)~~ Define software interrupt with example.
- ~~b)~~ What are the problems of linear decoding system in main memory array design? How to solve them?
- ~~c)~~ Interface two chips of 4k RAM and 1 chip of 2k ROM with processor. Starting address of RAM is 8000H.
2. ~~a)~~ What is Usability in HCI?
- ~~b)~~ Why is usability important in HCI? Give an example.
- ~~c)~~ Write down the Norman's Seven Principles for HCI design.

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Dept. of Computer and Communication Engineering
Faculty of Computer Science and Engineering
Patuakhali Science and Technology University
Dumki, Patuakhali-8602, Bangladesh

Final Examination of B. Sc. Engineering in CSE Level: 3 Semester: II Session: 2017-2018
Course Code CCE-321 Course Title Computer Peripheral and Interfacing July-December 2020 Credit: 03
Time: 03 Hr Marks: 70

Answer any 05 out of 06 Questions (Split answers are highly discouraged)

- 1 [A.] Enlist inside parts of optical mouse. How does an optical computer mouse work? 5
- [B.] What is Impact Printers? How does Impact Printers work? Explain with functional diagram. 6
- [C.] Draw the functional block diagram of 8251A PCI. 3
- [A.] What determines the quality of monitor? 3
- [B.] What is a RVDT? How does RVDT work step by step? Explain with RVDT construction diagram. 6
- [C.] What is RS-232 Handshaking? Explain with figure. 2
- [D.] Differentiate between span and range of a transducer system. 3
- [A.] a) What is a plasma monitor? 3
- b) What is nonlinearity error? 3
- [B.] Write down the difference between Sensors and Transforms. 3
- [C.] How does a process control loop work? Briefly explain with figure. 4
- [D.] Assume the ASCII character "A" (01000001), is framed in between the start bit and two stop bits. Notice that the LSB is sent out first. Now answer the following questions. 4
 - i. Define character-oriented transmission.
 - ii. How the data is packed?
 - iii. How many bits constitute the character?
 - iv. When the data begins and ends?
 - v. Distinguishes between Synchronous and Asynchronous Serial Communication.
 - vi. Which communication method is suitable for this transmission?
 - vii. Explain the character transmission with framing diagram.
- [A.] Show the pin configuration of 8255 PPI and describe the operation performed by each unit separately. 6
- [B.] Convert 39887.5625 to IEEE 32-bit floating point format. 4
- [C.] Explain software keyboard interfacing procedure with circuit connections and algorithm. 4
- [A.] Draw the architectural diagram of 8279 programmable keyboard/display controller. How does 8279 keyboard work? 6
- [B.] Differentiate between programmed I/O and interrupt initiated I/O. 4
- [C.] How can the signed binary numbers be represented? 3
- [D.] What is the range of signed values that may be stored in 20 bits? 1
- [A.] How to connect a microprocessor to ROM/RAM chips? Illustrate and explain each technique. 9
- [B.] When is an overflow flag set or reset? Give examples. 3
- [C.] What are the basic principles of two-wire handshaking method of data transfer? 2

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10
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116

Peripheral

Patuakhali Science and Technology University
Final Examination of B.Sc. Engg. (CSE), Level: 3 Semester: II
Course Code: CCE 321 Course Title: Computer Peripheral and Interfacing
Credit Hour: 3.00 Full Marks: 70 Duration: 03 Hours

[Figures in the right margin indicate full marks. **Split answering of any question is not recommended.** Write the full question number e.g. 2(B)(i) before the answer paragraph]

Answer any 5 of the following questions

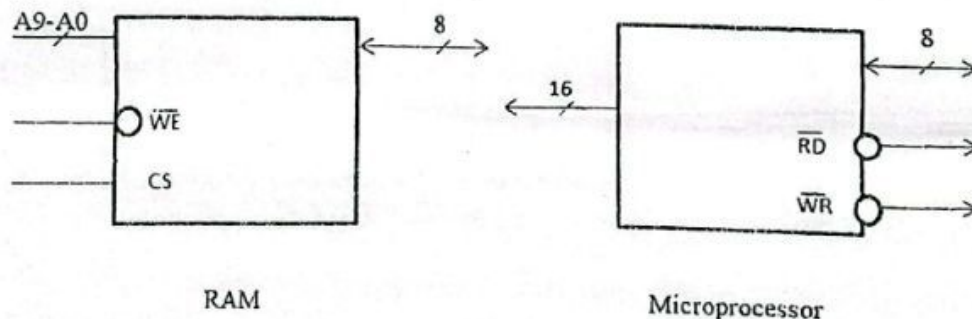
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|---|---|---|---|
| 1 | A | What is serial and parallel communication? Explain with example. | 4 |
| 1 | B | Define 8155 PPI. Write down the feature of 8155 PPI. | 5 |
| 1 | C | How does inkjet printer work? Explain with figure. | 5 |
| 2 | A | What is main difference between Sensors and Transforms? | 3 |
| 2 | B | (i) Define Programmable Logic Controllers System. | 2 |
| | | (ii) What is the purpose of opto isolators or optocoupler in a PLC? Explain with diagram of input /output channels. | 6 |
| 2 | C | What do you mean by impact and non-impact printer ? | 3 |
| 3 | A | What is an LVDT? How does LVDT work step by step? Explain with LVDT construction diagram. | 6 |
| 3 | B | What is <u>asynchronous communication protocol</u> ? Briefly describe RS-232C connection with a <u>simple example</u> . | 8 |
| 4 | A | Draw the functional block diagram of <u>8251A</u> PCI. | 3 |
| 4 | B | Briefly explain functional block diagram of 8251A PCI with consisting sections. | 7 |
| 4 | C | Which process of communication is controlled by an external circuit and also used for data transfer between memories and input/output ? Draw block diagram of a typical DMA controller. | 4 |
| 5 | A | (i) What is the IEEE 488 bus system? | 2 |
| | | (ii) Briefly explain the operation of IEEE 488 bus system based on the handshaking protocol with PIN configuration. | 6 |
| 5 | B | How does a paper scanner work? Explain with functional diagram. | 6 |
| 6 | A | <u>What is 82C55 PPI?</u> What will be the operation of stepper motor when it is connected with 82C55 PPI ? | 6 |
| 6 | B | How does laser printer work? Explain with functional diagram. | 6 |
| 6 | C | What is the example of biometric technology? | 2 |

6th Semester (L-3, S-II), Final Exam of B. Sc. Engg. (CSE), July -Dec. 2018
Course Code: CCE 321 Course Title: Computer Peripheral and Interfacing Session: 2015-2016
Credit Hour: 3.0 Full Marks: 70 Duration: 3 Hours

[Figures in the right margin indicate full marks. Split answering of any question is not recommended]

Answer any 5 of the following questions:

1. (a). Define transducers. Write down the advantages of electrical signal as output of a transducer. 4
(b). What do you mean by helipot? Describe the working procedure of L.V.D.T. 5
(c). Describe the effect of variation of the dielectric constant for the measurement of displacement in a capacitive transducer. 5
2. (a). Define piezo-electric effect. Derive the equation of voltage sensitivity due to piezo-electric effect. 5
(b). Write short notes on: i. Photoemissive cell ii. Photoconductive cell iii. Photovoltaic cell. 3
(c). What is hall effect? Describe the working principle of hall effect transducers. 4
(d). Write down the applications and limitations of thermistors. 2
3. (a). Define pyrometer. Describe the working principle of radiation pyrometer. Write down the advantages and limitations of radiation pyrometer. 6
(b). What is thermocouple? Briefly discuss about the construction and operating principle of a thermocouple. 5
(c). Write short notes on: i. U-type manometer ii. Well-type manometer. 3
4. (a). Describe Data Read and Data Write operations through FDC. 5
(b). Encode data 101000100110 using FM, MFM and RLL encoding formats. Show the waveforms. 9
5. (a). Assume the following RAM chip and the microprocessor: 6



Connect the microprocessor to the RAM to obtain the memory map: 0000_{16} through $0FFF_{16}$. Use only the signals shown. Draw a neat logic diagram and analyze the memory map.

- (b). The MMU can perform address translation in one of the two ways. Define them. 3
- (c). Perform binary addition of $(-66)_{10}$ and $(-65)_{10}$. Now determine the carry, sign, overflow, and zero flags for this addition. 5
6. (a). What are the basic differences between ASCII and EBCDIC? 2
- (b). Explain the operation of a delta-delta shadow-mask CRT. 3
- (c). What do you mean by true color representation? Does it represent any disadvantages? 3
- (d). Write short notes on: 6
- Interlaced refresh procedure
 - 10H8
 - Bitmap and Pixmap

2016

16

Patuakhali Science and Technology University
6th Semester (L-3, S-II), Final Exam of B. Sc. Engg. (CSE), July -Dec. 2016
Course Code: CCE 321 Course Title: Computer Peripheral and Interfacing Session: 2013-2014
Credit Hour: 3.0 Full Marks: 70 Duration: 3 Hours

[Figures in the right margin indicate full marks. Split answering of any question is not recommended]

Answer any 5 of the following questions:

2016

1. (a). Differentiate between MDA and CGA. 4
(b). Write down the advantages and disadvantages of LCD over CRT display. 4
(c). When a key on a keyboard is pressed, a switch is activated to generate signals for communicating the same to the PC. Give a brief description of four of them. 8
2. (a). What is key debouncing? Why is it needed? 2
(b). Define make code, break code, typematic, persistence, aspect ratio and resolution. 3
(c). Modern keyboards support bidirectional data transfer. How? 5
(d). Explain the phase sequences for the execution of "Test Unit Ready" SCSI command. 4
3. (a). Why is serial ATA used in present technologies instead of parallel ATA? 3
(b). Describe IDE command execution protocols. 10
(c). What do you mean by "CHRN" value? 1
4. (a). Describe Data Read and Data Write operations through FDC. 5
(b). Encode data 101000100110 using FM, MFM and RLL encoding formats. Show the waveforms. 9
5. (a). Illustrate an 8-bit microprocessor interfaced to a 4K RAM system using a full/partial decoded addressing technique. Also show the address map corresponding to the above mentioned organization. 5
(b). What do you mean by ISA bus and PCI bus? 2
(c). Write down the steps involved in starting a printing sequence using a printer controller chip. 4
(d). Explain the operation of a delta-delta shadow-mask CRT. 3
6. (a). What are the basic differences between ASCII and EBCDIC? 2
(b). Describe the process of a device initiated I/O transfer. 3
(c). When is an overflow flag set or reset? Give examples? 3
(d). Illustrate the character and signal generation procedure of a CRT display with a short description. 4
(e). The MMU can perform address translation in one of the two ways. Define them. 2

Memory

Patuakhali Science and Technology University

B.Sc.Engg.(CSE) L-3, S-2, Final Examination -2015 (July-Dec). Session: 2012-13

Course Code: CCE-321

Course Title: Computer Peripheral and Interfacing

Credit Hour: 3.0

Full Marks: 70

Duration: 3 hours

[Figures in the right margin indicate full marks. Split answering of any question is not recommended]

[Use figures where necessary]

Answer any 5 of the following questions:

P. 50

1. (a). What is buffer? Why we use buffers? 3
(b). Define interfaces. What are the characteristics of computer peripherals? 4
(c). Differentiate between buffers and spoolers. 2
(d). What do you mean by solid state storage device? Write down the features and advantages of Solid state storage devices. 5
2. (a). Define raster scan display and random scan display. What are the differences between raster scan system and random scan system? 5
(b). What are the advantages of LCD panel over conventional CRT monitor? 4
(c). Differentiate between ATA RAID and SCSI RAID. 2
(d). Write down the selection criteria of a monitor. 3
3. (a). Define USB. What are the functions of USB ports? Write down the limitations of USB ports. 5
(b). Differentiate between IEEE1394 and USB. 3
(c). Sketch the three tier topology of USB. Write down the benefits of three tier topology of USB. 4
(d). What are the benefits of using USB? 2
4. (a). What is PCI Express? Write down the features and applications of PCI Express bus. 7
(b). Define resource conflict. How can user resolve resource conflict? 4
(c). Differentiate between serial ATA and parallel ATA. 3
5. (a). Differentiate between ISA, VESA and EISA buses. 5
(b). What do you mean by ATA? Write down the features of the ATA-1 specification. 4
(c). Write short note on: AGP, Plug and Play, SCSI. 3
(d). Differentiate between IRQ and DMA channels. 2
6. (a). Define RAID. Describe the different levels of RAID. 7
(b). Differentiate between IDE/ATA and SCSI. 4
(c). Write down the advantages of SCSI over ATA. 3

Patuakhali Science and Technology University

B.Sc. Engg. (CSE) L-3, S-2, Final Examination -2014 (July-Dec), Session: 2011-12
Course Code: CCE-321 Course Title: Computer Peripheral and Interfacing
Credit Hour: 3.0 Full Marks: 70 Duration: 3 hours

[Figures in the right margin indicate full marks. Split answering of any question is not recommended]
[Use figures where necessary]

Answer any 5 of the following questions:

1. (a). What is buffer? Why we use buffers? 4
(b). Define interfaces. What are the characteristics of computer peripherals? 4
(c). What do you mean by solid state storage device? Write down the features and advantages of Solid state storage devices. 6
2. (a). Define raster scan display and random scan display. What are the differences between raster scan system and random scan system? 6
(b). What are the advantages of LCD panel over conventional CRT monitor? 5
(c). Differentiate between IRQ and DMA channels. 3
3. (a). Define USB. Write down the limitations of USB ports. 5
(b). What are the benefits of using USB? 5
(c). Sketch the three tier topology of USB. Write down the benefits of three tier topology of USB. 4
4. (a). What is PCI Express? Write down the features and applications of PCI Express bus. 8
(b). Define resource conflict. How can user resolve resource conflict? 4
(c). What do you mean by voltage conversion and protocol conversion? 2
5. (a). Compare between VESA and PCI buses. 5
(b). What do you mean by ATA? Write down the features of the ATA-1 specification. 6
(c). Write short note on: AGP, Plug and Play, SCSI. 3
6. (a). Define RAID. Describe the different levels of RAID. 7
(b). Differentiate between IDE/ATA and SCSI. 4
(c). Write down the advantages of SCSI over ATA. 3

Patuakhali Science and Technology University

Faculty of Computer Science and Engineering

Dept. of Computer and Communication Engineering

Final Examination of B. Sc. Engineering In CSE Level: 3 Semester: II Session: 2018-2019

Course Code
CCE-322

Course Title
Computer Peripheral and Interfacing Sessional

July December 2021

Credit: 1.50
Marks: 70

1. Answer the following:

2*10

- a. Illustrate the Gantt chart for your project.
- b. What is the function of the IOREF pin on the Arduino UNO?
- c. Arduino IDE consists of 2 functions. Define them.
- d. What is the use of the Arduino.h header file?
- e. What is the execution process of an Arduino code?
- f. Why does every compilation of a source code in Arduino check for previous compilations?
- g. What is the use of the ESP8266 WiFi Module?
- h. What is the way of throwing an error using preprocessing directives to the Arduino Compiler and forcing it to stop compilation?
- i. What will happen if we supply a voltage of 25V to the Vcc of the Nokia 5110 GDM?
- j. What is the output of "pin1" if "pin2" is sent "1011" where 1 is 5V and 0 is 0V?

```
1. int pin1 = 12;  
2. int pin2 = 11;  
3. void setup() {  
4.     pinMode(pin1, OUTPUT);  
5.     pinMode(pin2, INPUT);  
6.     Serial.begin(9600);  
7. }  
8. void loop() {  
9.     if(digitalRead(pin2)==1) {  
10.         digitalWrite(pin1,LOW);  
11.     }  
12.     else if(digitalRead(pin2)==0)  
13.         digitalWrite(pin1,HIGH);  
14.     }  
15. }
```

2. Make the project report.
3. Develop and display the Project.
4. Attend viva-voce.

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20 (5)

Prof. Dr. Md. Rasheduzzaman
Dept. of Computer and Communication Engineering
Patuakhali Science and Technology University
Patuakhali, Bangladesh

Dept. of Computer and Communication Engineering
Patuakhali Science and Technology University

6th Semester (Level-3, Semester-II), Midterm Examination of B.Sc. Engg. (CSE), July-December: 2023

Course Code: CCE-321 Course Title: Computer Peripheral and Interfacing

Credit Hour: 3.0 Full Marks: 15 Duration: 60 Minutes

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|---|----|---|-----|
| 1 | a) | Differentiate between standard I/O and memory mapped I/O. | 2 |
| | b) | What are the main types of interrupt? Describe with example. | 2 |
| | c) | Interface two chips of 4k RAM and two chips of 2k ROM with processor. Starting address of RAM is 9000H. | 3.5 |
| 2 | a) | Explain the different communication models used in IoT. How do these models impact data transmission, scalability, and energy efficiency in IoT systems? | 3.5 |
| | b) | What does it mean by Internet of Underground Things and Internet of Underwater Things? | 2 |
| | c) | The Internet of Things (IoT) is often described in three key dimensions. Explain each dimension with relevant examples. How do these dimensions collectively enable a seamless IoT ecosystem? | 2 |

Dept. of Computer and Communication Engineering
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Dumki, Patuakhali-8602, Bangladesh

Final Examination of B. Sc. Engineering in CSE Level: III Semester: II Session: 2020-2021

Course Code	Course Title	July-December 2023	Credit: 03
CCE-321	Computer Peripheral and Interfacing		Time: 03 Hr
			Marks: 70

Answer any 05 out of 06 Questions (Split answers are highly discouraged)

- 1 [A.] Which design features are essential in WSNs to optimize data collection, communication, and energy efficiency? 3 (2)
- [B.] Explain the structure of a potentiometer sensor used for measurement of linear displacement with electric circuit diagram and proper equation. List the applications of potentiometer sensor in/around your home and office/university. 4
- [C.] Write down the primary objective and idea of Timeout-MAC in wireless sensor networks. How does T-MAC differ from S-MAC in terms of energy efficiency and what mechanism does T-MAC use to reduce idle listening in sensor nodes? 4
- [D.] Define proximity switches. Sketch the configurations of different contact-type proximity switch being used in manufacturing automation. Explain in brief two applications of "Reed switch". 3 (2)
- 2 [A.] 'Bellows are more sensitive than capsules in Fluid pressure sensor'. State true or false and justify your answer. 3
- [B.] How does a capacitive sensor operate to measure linear displacements, and what are the typical applications where such non-contact sensors are preferred over other types of displacement sensors? Explain all with appropriate diagram. 4
- [C.] Write down about Hidden Terminal Problem and Exposed Terminal Problem for the CSMA MAC-Protocol in wireless sensor network. Explain with suitable figure. 4
- [D.] Briefly explain the following IoT data protocols with their operation. 3
 - i. MQTT
 - ii. M2M
- 3 [A.] What do you mean by nonlinearity error? How it is different than Hysteresis error? (2)
- [B.] How does a Hall effect sensor detect the presence and strength of a magnetic field? Explain the principle of working of Hall effect sensor. 3
- [C.] What is a LVDT? How does LVDT work step by step? Explain with LVDT construction diagram. 5 (5)
- [D.] How does Slotted ALOHA improve upon the basic ALOHA protocol in terms of efficiency? Explain with frames transition in a pure ALOHA and a Slotted ALOHA network. 4
- 4 [A.] Draw the architectural diagram of 8279 programmable keyboard/display controller. How does 8279 keyboard work? (5)
- [B.] Interface two chips of 4k RAM and two chips of 2k ROM with processor. Starting address of RAM is A000H. Draw a neat logic diagram of your implementation. 4
- [C.] Given a memory with a 14-bit address and an 8-bit word size. 3
 - i. How many bytes can be stored in this memory?
 - ii. If this memory were constructed from 1K x 1 RAMS, how many memory chips would be required?
 - iii. How many bits would be used for chip select?
- [D.] Define the three types of I/O. Identify each as either microprocessor-initiated or device-initiated. 2
- 5 [A.] Show the software keyboard interfacing process through a flow chart. Describe each step for a 4x4 keyboard. 5

[B.] Consider the hardware schematic shown in Figure 01.

- Determine the address map of this system. Note: $\overline{\text{MEMR}} = 0$ for read, $\overline{\text{MEMR}} = 1$ for write, $\overline{\text{MIO}} = 0$ for I/O and $\overline{\text{MIO}} = 1$ for memory.
- Is there any possibility of bus conflict in this organization? Clearly justify your answer.

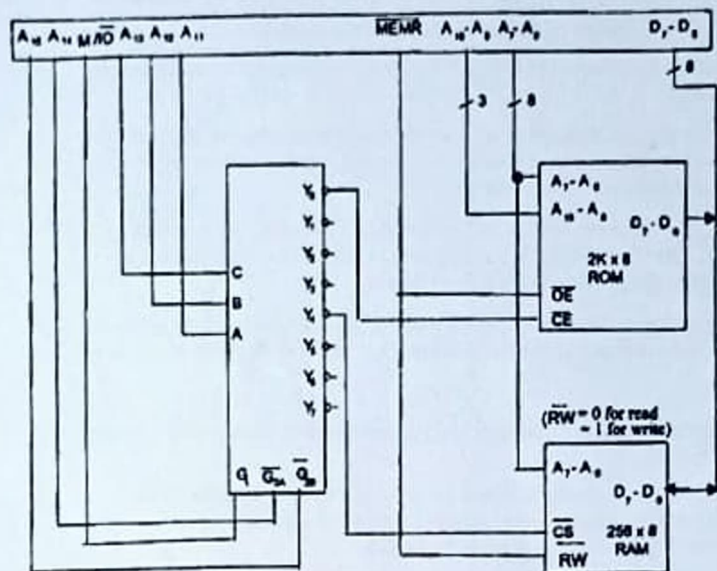


Figure: 01

[C.] Summarize the basic difference between maskable and nonmaskable interrupts. Describe how power failure interrupt is normally handled.

[D.] i. What is fluorescence?

- If we use 2-byte pixel values in a 24-bit lookup table representation, how many bytes does the lookup table occupy?

6 [A.] The block diagram of a 512×8 RAM chip is shown in Figure 02. In this arrangement the memory chip is enabled only when $\overline{\text{CS1}} = \text{L}$ and $\text{CS2} = \text{H}$. Design a $1\text{K} \times 8$ RAM system using this chip as the building block. Draw a neat logic diagram of your implementation. Assume that the microprocessor can directly address 64K with a $\text{R}/\overline{\text{W}}$ and eight data pins. Using linear decoding and don't-care conditions as 1's, determine the memory map in hexadecimal.

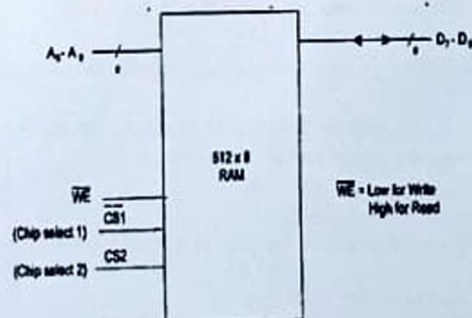


Figure: 02

[B.] Serialize the steps associated in turning on or off a pixel in LCD monitor.

[C.] List the methods used in main memory array design. What are the advantages and disadvantages of each?

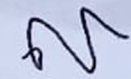
[D.] Obtain the control word when the ports of 8255A are to be used in mode 0 with port-A as output port and port B as input port and port C as output port.

[E.] An interface incorporates several components including its communication type as well as the interface software. List out eight major communication types in case of a printer.

Dept. of Computer and Communication Engineering
Faculty of Computer Science and Engineering
Patuakhali Science and Technology University

Final Examination of B. Sc. Engineering in CSE Level: III Semester: II Session: 2020-2021

Course Code	Course Title	July-December 2023	Credit: 1.50
CCE-322	Computer Peripheral and Interfacing sessional		Time: 03 Hr
			Marks: 70

- | | | | |
|----|--|--|----|
| 1. | Viva | | 20 |
| 2. | Write down <u>project title, objectives.</u> | | 5 |
| 3. | Circuit diagram/ flow chart / equipment | | 5 |
| 4. | Project | | 30 |
| 5. | Report | | 5 |
| 6. | Continuous assessment | | 5 |
- 

Dept. of Computer and Communication Engineering
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Patuakhali Science and Technology University
Dumki, Patuakhali-8602, Bangladesh

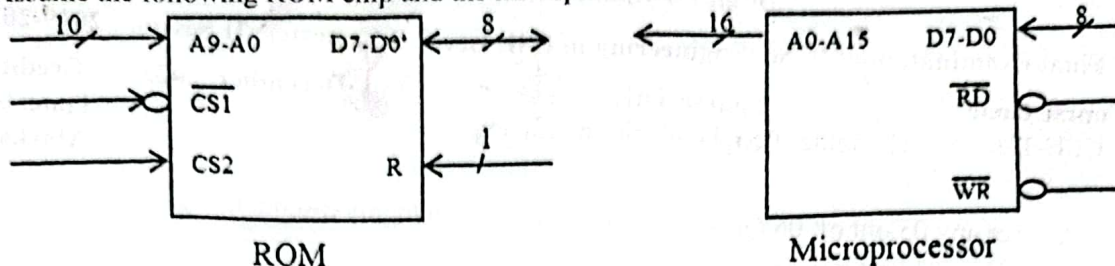
Final Examination of B. Sc. Engineering in CSE Level: 3 Semester: II Session: 2019-2020

Course Code	Course Title	July-December 2022	Credit: 03
CCE-321	Computer Peripheral and Interfacing		Time: 03 Hr
			Marks: 70

Answer any 05 out of 06 Questions (Split answers are highly discouraged)

1. ☒ [A.] What are the different components of IoT? How does the Internet of Things (IoT) affect our everyday lives? 4
- ☒ [B.] Explain the structure of a potentiometer sensor used for measurement of linear displacement with electric circuit diagram and proper equation. List the applications of potentiometer sensor in/around your home and office/university. 5
- ☒ [C.] What is Human Computer Interaction? Briefly explain most important components of HCI. 5
2. [A.] What are the different types of IoT security measures, and what are some effective strategies or best practices for securing IoT devices to protect them from potential threats and vulnerabilities? 3
- [B.] How does a capacitive sensor operate to measure linear displacements, and what are the typical applications where such non-contact sensors are preferred over other types of displacement sensors? Additionally, discuss the range of displacements that capacitive sensors are capable of measuring and the factors that affect their accuracy. Explain all with appropriate diagram. 4
- [C.] Imagine you're designing a real-time system for a futuristic smart city where various subsystems need to interact seamlessly to ensure optimal performance. In this context, how would you incorporate both periodic and aperiodic task models to manage different types of operations? 4
- [D.] Write down Norman's Seven Principles. 3
3. ☒ [A.] Define nonlinearity error. How it is different than Hysteresis error? 3
- ☒ [B.] What are the key components and characteristics of a digital ecosystem, and how do these elements interact to create value within industries such as finance, healthcare, and e-commerce? Additionally, discuss the challenges and opportunities associated with managing and sustaining a digital ecosystem in a rapidly changing technological environment. 3
- ☒ [C.] What is a LVDT? How does LVDT work step by step? Explain with LVDT construction diagram. 5
- [D.] What are the definitions and characteristics of active and passive transducers, and how do they differ from each other? Additionally, what are the key distinctions between sensors and transducers in terms of their functions and applications? 3
4. ☒ [A.] Show the pin configuration of 8255 PPI and separately describe the operation performed by each unit. 5
- [B.] Illustrate the key components of a laser printer and provide a brief description of the function each component performs. 4
- ☒ [C.] Convert -1313.3125_{10} to IEEE 32-bit floating point format. 3
- ☒ [D.] Mention the main functions of an interface. 2

5 [A.] Assume the following ROM chip and the microprocessor:



Connect the microprocessor to the ROM to obtain the memory map: 0000₁₆ through 03FF₁₆.

Use only the signals shown. Draw a neat logic diagram and analyze the memory map.

[B.] When you press a key on your computer, you are activating a switch. There are many different ways of making these switches. Describe any four of them.

[C.] Explain in detail the working principle of an LCD monitor and how its components interact to produce images on the screen?

[D.] What is key debouncing? Why is it needed?

- [A.] i. If we want to cut a 512 x 512 sub-image out from the center of an 800 x 600 image, what are the coordinates of the pixel in the large image that is at the lower left corner of the small image?
- ii. Find the CMY coordinates of a color at (0.2, 1, 0.5) in the RGB space.
- iii. If we use direct coding of RGB values with 2 bits per primary color, how many possible colors do we have for each pixel?
- iv. What is the pitch of a color CRT?
- v. Why do many color printers use black pigment?

[B] How can we assign input and output port of 8255?

[C] In USB communication, a single transaction involves the transmission of up to three packets. Mention the names of these packets and describe the specific services each one provides during the communication process.

[D] Provide a detailed comparison of Random Scan Displays and Raster Scan Displays, focusing on their operational mechanisms, image rendering techniques, applications, and overall performance characteristics.

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Dumki, Patuakhali-8602, Bangladesh

Final Examination of B. Sc. Engineering in CSE Level: 3 Semester: II Session: 2018-2019

Course Code
CCE-321

Course Title
Computer Peripherals and Interfacing

July-December 2021

Credit: 03

Time: 03 Hr

Marks: 70

- Answer any 05 out of 06 Questions (Split answers are highly discouraged)
1. [A.] What are the different components of IoT? How does the Internet of Things (IoT) affect our everyday lives? 4
 - [B.] What is communication model in IoT? Explain different types of communication model in IoT with diagram and example. 6
 - [C.] Write down the difference between Embedded and Real Time Systems. 4
 2. [A.] What is the periodic and aperiodic task model in real-time system? Explain with example. 4
 - [B.] What is a RVDT? How does RVDT work step by step? Explain with RVDT construction diagram. 4
 - [C.] How many principles are there in Norman design principles? Write down Norman's design principles for IoT. 3
 - [D.] Briefly explain the most 6 communication protocols used by IoT. 3
 3. [A.] Explain the characteristics of RTS. 3
 - [B.] Write down the difference between Sensors and Transducer. 3
 - [C.] What is Human Computer Interaction? Briefly explain most important components of HCI. 5
 - [D.] What are the types of IoT security? How to Secure IoT Devices? 3
 4. [A.] a. What do you mean by true color representation? Does it represent any disadvantages? How to handle the problems? 5
 - b. If we use 5 bits each for red and blue and 6 bits for green in direct coding method, how many possible simultaneous colors do we have?
 - c. Interpret EFE_{16} as unsigned, signed, and floating point numbers that a microprocessor can understand. 4
 - d. What are the common problems with scanner? Share some tips to troubleshoot the basic scanner related problems. 3
 - e. Obtain the control word when the ports of 8255A are to be used in mode 0 with port-A as output port and port B as input port and port C as output port. 2
 5. [A.] How do the USB peripherals communicate with microprocessor? Give the pin-out for USB type C 3.0. 5
 - [B.] Categorize and explain the operational modes of 8279. How many ways the keyboard is interfaced with the CPU? 4
 - [C.] The microcomputer communicates with an external device via one or more registers called I/O ports using programmed I/O. Illustrate the functions of two of them with brief description. 3
 - [D.] Make a comparison between buffers and spoolers. 2
 6. [A.] Suppose, a processor is having 16 bits address line and it is byte addressable. Interface 4096x8 RAM with the processor. 5
 - a. What is the memory capacity of the processor?
 - b. How many address lines are required for a RAM?
 - c. How many RAM chips are required?
 - d. What will be the size of address decoder?
 - [B.] Show the software keyboard interfacing process through a flowchart. Describe each step for a 4x4 keyboard. 4
 - [C.] Serialize the steps associated in turning on or off a pixel in LCD monitor. 3
 - [D.] An interface incorporates several components including its communication type as well as the interface software. List out eight major communication types in case of a printer. 2

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